Project Design Phase
Solution Architecture

Date	07/03/2026
Team ID	SWTID-2026-6411
Project Name	product inventory management
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for the Product Inventory Management System focuses on building a secure, scalable, and maintainable backend using Node.js, Express.js, and MongoDB. The backend provides RESTful APIs for authentication and product management while ensuring data privacy through JWT-based authorization.

Goals of the Solution Architecture:

Identify the Best Tech Solution:

- Use Node.js, Express.js, MongoDB, and JWT authentication to build a secure and efficient backend API for product inventory management.

Define Structure & Characteristics:

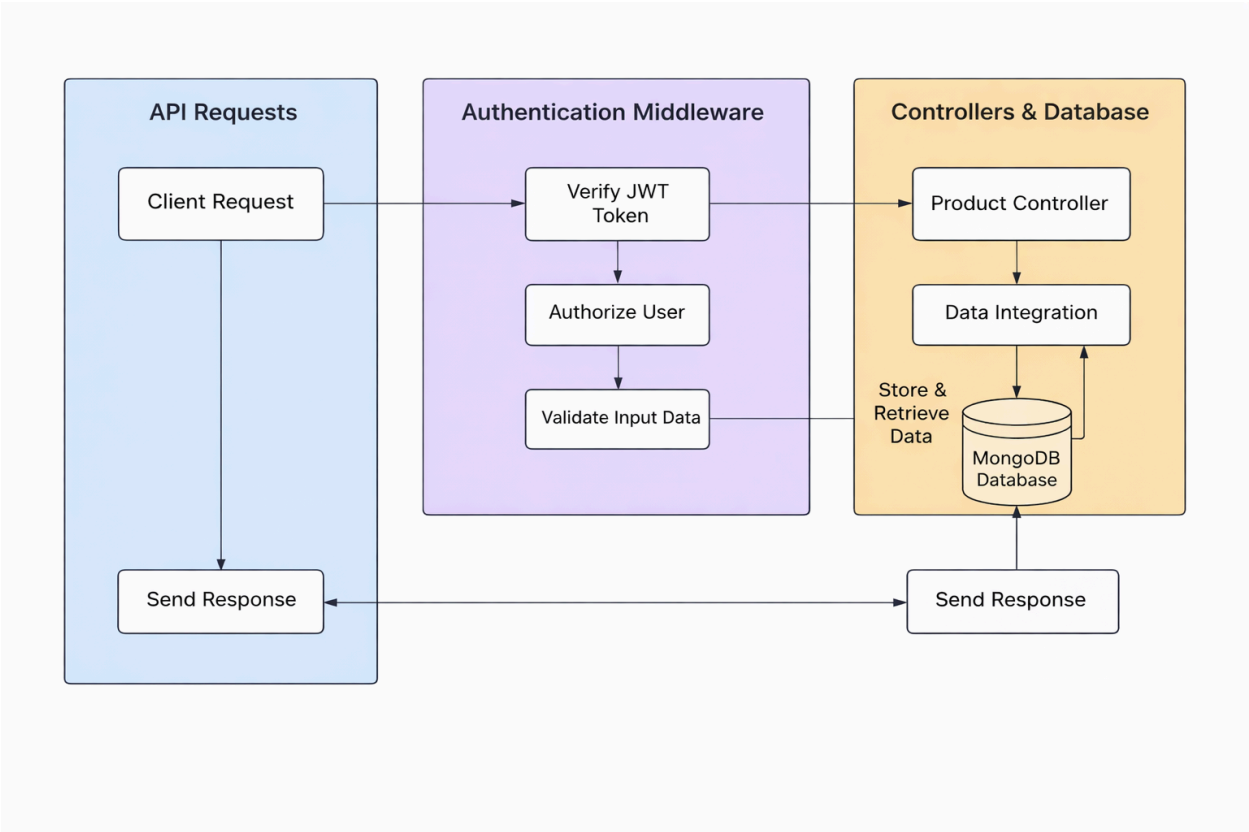
- Follow a modular MVC architecture to ensure scalability, maintainability, and clear separation of authentication and product management logic.

Outline Features & Development Phases:

- Plan development in phases including backend setup, authentication APIs, product CRUD operations, and testing for stable delivery.

Establish Specifications for Development & Delivery:

- Define REST API standards, token-based authorization, and structured middleware to maintain consistent and secure data flow.
- Optimize database queries and asynchronous API handling to provide fast response times and reliable backend performance.


API Requests
Client Request
Send Response
Authentication Middleware
Verify JWT Token
Authorize User
Validate Input Data
Controllers & Database
Product Controller
Data Integration
Store & Retrieve Data
MongoDB Database
Send Response